Beneath the Surface: Catching Cartels in the Act in Brazilian Procurement*

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Abstract

This study leverages convictions from Brazil's antitrust authority (CADE) as ground truth to validate a Regression Discontinuity design inspired by Kawai et al. (2023) and a portfolio of complementary detection screens, applied to 3.8 million public procurement auctions conducted in São Paulo, Brazil. By comparing narrowly winning and losing bids within auctions involving convicted cartel members, we identify statistically significant signals across multiple screen families—but with magnitudes and signs that diverge sharply from those predicted by canonical bid-rotation theory. Network-based screens (worker-flow and co-bidding overlap) substantially outperform classical variance and incumbency screens, achieving AUCs above 0.80 against AUCs near chance for variance-based detectors. Our findings highlight that in developing countries, anti-competitive behaviors—even when unrelated to explicit corruption—generate detection signatures distinct from those documented in advanced economies, with direct implications for the design of antitrust screening programs in emerging markets.

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1. Introduction

Bid rigging is among the most costly forms of anti-competitive behavior in public procurement: estimates from exposed cases place the average cartel overcharge between 15 and 30 percent of contract value, and the OECD conservatively attributes to collusion an inefficiency wedge on the order of one fifth of public purchases in affected markets (Organisation for Economic Co-operation and Development, 2015; Bosio et al., 2022). Because explicit evidence of coordination is almost always absent from the administrative record, a growing literature has sought to infer collusion indirectly from the statistical properties of bidding data. Classical detection screens exploit the fact that cartel agreements compress the distribution of bids: colluding firms submit prearranged prices that cluster tightly around the designated winner, yielding low bid dispersion, few active bidders, and predictable rotation patterns (Porter and Zona, 1993; Pesendorfer, 2000; Bajari and Ye, 2003; Conley and Decarolis, 2016; Imhof et al., 2018). A complementary and influential approach, recently advanced by Kawai et al. (2023), focuses on the threshold between winning and losing: under competition, the observable characteristics of narrowly winning and narrowly losing firms should be continuous at a bid margin of zero, and any statistically sharp discontinuity at the cutoff identifies an anti-competitive wedge. Applied to Japanese construction procurement, this Regression Discontinuity (RD) design uncovers sharp compression in bidder backlog and strong incumbency at the cutoff—signatures consistent with the canonical model of rotation-based collusion.

A first-order question for this literature is whether its screens travel. The cumulative evidence supporting canonical screen predictions is drawn almost exclusively from cartels operating in high-income economies with deep formal labor markets, dense bidder pools, and strong institutional capacity for antitrust enforcement. Whether the same statistical signatures appear in cartels operating in thinner institutional environments is an open question—and one with direct policy stakes, because emerging-market competition authorities increasingly rely on off-the-shelf screening tools developed and validated abroad to triage investigations (Organisation for Economic Co-operation and Development, 2015; Bosio et al., 2022). If canonical screens systematically mislabel cartels under these conditions, existing enforcement pipelines may be misfiring in precisely the settings where deterrence capacity is weakest.

This paper leverages a rare opportunity to answer that question directly. Between 2009 and 2019, Brazil's federal antitrust authority—the Conselho Administrativo de

Defesa Econômica (CADE)—prosecuted a series of procurement cartels operating in the state of São Paulo, securing administrative convictions against 35 firms across seven distinct cartel cases, with case files identifying the specific product markets and time windows over which each ring was active. We match these convicted firms, and the tender windows and product markets delineated in CADE's administrative records, to the universe of 3.8 million individual bids placed on São Paulo's electronic procurement platform, the *Bolsa Eletrônica de Compras* (BEC), over the same period. We further link the resulting bid-level dataset to the universe of formal-sector employment relationships in Brazil through the Relação Anual de Informações Sociais (RAIS), enabling us to characterize the workforce composition and worker mobility histories of every firm in the BEC at the time of each bid. To our knowledge, this is the first cartel-detection study to combine auction-level behavioral data, employer–employee matched labor records, and formal antitrust convictions as a single integrated benchmark.

Our research design exploits two complementary benchmark samples. The first is a *pair-matched* sample of 107 pregão auctions in which CADE rulings directly implicate both the winning and the runner-up firms as co-conspirators in the same cartel case during the same active window—the cleanest ground truth available for any large-scale procurement-cartel study to date. The second is a *cartel-active* sample of roughly 2,350 auctions in which at least one convicted firm participated during the active period of its cartel. The pair-matched sample delivers precision at the cost of statistical power; the cartel-active sample delivers power at the cost of a noisier treatment. Reporting results in parallel across both benchmarks disciplines our inference: a claim that survives both lenses is credible, a claim that appears in only one demands interpretation. We refer to the two samples collectively as *conviction-anchored detection benchmarks*.

Our first set of results documents a striking failure of canonical screens in this institutional setting. On the pair-matched sample of 107 auctions, classical variance screens reverse their expected sign: relative to the population of comparable clean auctions, bid spread, the coefficient of variation of bids, and the number of active bidders are all *higher*—not lower—at cartel-tainted auctions, with two-sided p-values below 0.02 in every case. These inversions are the opposite of the compression predicted by the canonical bid-rotation model that motivates classical screens and underpins Kawai et al. (2023). On the broader cartel-active sample, where power is not a binding constraint, the same classical screens recover their canonical sign but lose virtually all discriminative power: the areas under the receiver-operating-characteristic curves (AUCs) for bid

spread, coefficient of variation, and bidder count all lie between 0.46 and 0.48, statistically indistinguishable from chance. A direct replication of the Kawai close-margin RD on the cartel-active subsample yields no statistically detectable discontinuity in incumbency or last-bid behavior at the cutoff, and a density test in the spirit of Imhof et al. (2018) detects no manipulation of the running variable. In short, neither the canonical variance-based screens nor the RD screen of Kawai et al. (2023) detect the cartels that CADE has formally convicted.

Our second set of results shows that network-based screens constructed from administrative labor data recover strong and statistically robust cartel detection on the same benchmarks. For each auction, we construct two network screens from the five-year window of RAIS records preceding the tender: the number of workers who have moved between the winner and the runner-up firms, and the Jaccard similarity of the two firms' historical workforces. These yield AUCs of 0.825 and 0.815 respectively on the cartel-active benchmark, outperforming every classical screen by wide margins. A third screen, built from the co-bidding network of shared tender participations, yields an AUC of 0.69, providing independent corroboration through a distinct information channel. Critically, the worker-flow signal is neither an artifact of sectoral composition nor a mechanical re-expression of co-bidding intensity. Restricting attention to firm pairs that share the same 2-digit sectoral classification leaves the worker-flow AUC essentially unchanged at 0.8249, and the residual of worker-flow after partialling out the co-bidding density of each pair still achieves an AUC of 0.74. The conviction-anchored benchmark does not merely declare cartels detectable in São Paulo; it pinpoints the channel through which the detection signal propagates—worker mobility, not bid dispersion.

We interpret these findings through the lens of institutional thinness. In settings where the universe of firms capable of bidding on a given product is small and the formal-sector labor market for the specialized workers that serve those bids is shallow, coordination among colluding firms generates naturally detectable traces in worker flows: personnel rotations, key-employee transfers, and shared specialized labor become vehicles for, and therefore signatures of, the collusive communication that substitutes for the statistical-pattern coordination captured by canonical screens in thicker markets. This mechanism—which we develop formally in a simple model in Section 4—rationalizes both the failure of bid-dispersion screens and the success of network-based screens in the Brazilian setting, and it predicts that classical screen-based enforcement strategies will systematically under-detect cartels in precisely the institutional environments where

their deterrence value would be highest.

This paper makes three contributions. *First*, methodologically, we provide the first stress test of the RD cartel-detection framework of Kawai et al. (2023) against an independent criminal ground truth, and we document that its canonical predictions can reverse sign in institutional environments with thin bidder pools. This is not a repudiation of the close-margin RD design, which remains internally valid; rather, it is a warning that the *direction* of its test statistic depends on primitives that are not invariant across settings. *Second*, substantively, we introduce conviction-anchored detection benchmarks by linking CADE rulings, BEC bid-level records, and RAIS employer–employee data, and we show that worker-flow and co-bidding network screens constructed from these sources substantially outperform classical screens in detecting convicted cartels. Because such data are increasingly available to national competition authorities in middle-income economies, the screens we propose are directly deployable at negligible marginal cost. *Third*, conceptually, we advance the view that collusion leaves institution-specific signatures, and that the design of an effective detection screen should depend on which coordination technology is cheapest for cartel members to operate in the relevant environment—a framing that organizes and partially reconciles the growing patchwork of contradictory empirical findings in the detection literature (Conley and Decarolis, 2016; Imhof et al., 2018; Kawai et al., 2023).

The remainder of the paper is structured as follows. Section 3 describes the institutional setting of São Paulo's public procurement and the CADE cartel cases that anchor our ground truth. Section 2 situates our contribution in the procurement-collusion detection literature and in the adjacent literatures on worker mobility and law-and-economics enforcement. Section 5 describes the BEC, RAIS, and CADE data sources and the construction of the pair-matched and cartel-active benchmark samples. Section 4 presents our empirical strategy—the RD replication, the portfolio of classical screens, and the network-based screens—and sketches a simple theoretical framework linking screen predictions to institutional primitives. Section 6 reports the main results. Section 8 discusses policy implications and concludes.

2. Literature Review

Collusion in public procurement auctions has been extensively studied through the lenses of game theory, empirical analysis, and policy evaluation. A rich theoretical

literature explains how bidders can sustain collusive agreements in repeated auction environments. Key mechanisms include bid rotation, where cartel members take turns winning contracts, and strategies to deter outsider entry. For example, dynamic models show that bidding rings can allocate wins intertemporally without side payments by rotating bids in a predetermined order, ensuring that each cartel member secures a share of contracts over time. Auction format and information flow also matter: tacit coordination can emerge even without explicit communication, especially under auction designs that facilitate monitoring of rivals' behavior. Notably, uniform-price auctions tend to relax competitive pressure more than pay-as-bid formats, thereby facilitating implicit collusion (Porter and Zona, 1993).

Empirical evidence since the 2000s demonstrates that collusive schemes are not merely theoretical constructs but real phenomena observed across different regions. Numerous documented cartels in procurement markets illustrate various collusive strategies. In Japan, for instance, construction contractors have systematically rigged bids on public works projects, with nationwide tender data revealing widespread coordination (Kawai et al., 2023). Within such schemes, firms often allocate subsequent contracts to the member that has gone the longest without winning, a clear indicator of bid rotation. Similar patterns have been identified in Italy, Scandinavian countries, and in Brazil's public contracting, where studies document that collusive rings inflate contract prices and coordinate bidding behavior through predictable bid rotations and artificially close bids (Bajari and Ye, 2003). These patterns persist over time, leading to higher procurement costs and diminished competition (Conley and Decarolis, 2016).

Given the covert nature of collusion, researchers have focused on developing robust detection methods. Traditional econometric approaches leverage the insight that collusion distorts the statistical properties of bids. For example, tests have been developed to assess whether bidders' cost estimates or bid rankings are exchangeable under competitive conditions, with systematic deviations serving as red flags for collusion (Bajari and Ye, 2003). More recent work has advanced these methods by applying Regression Discontinuity (RD) designs that exploit near-tie auctions, where the winning margin is nearly zero. Under true competition, firms in such close contests should display similar characteristics, and any systematic differences can indicate collusive coordination (Imhof et al., 2018). These techniques have successfully detected collusion in various procurement settings, highlighting that even sophisticated cartels leave measurable traces in bidding behavior.

Policy responses to collusion have evolved significantly since 2000. Regulatory frameworks now emphasize proactive detection and robust enforcement. Leniency programs, which incentivize cartel members to self-report by offering reduced penalties, have been widely implemented and have been shown to destabilize collusive arrangements (Organisation for Economic Co-operation and Development, 2015). In parallel, governments have refined auction design—by increasing transparency, enforcing strict bid submission protocols, and adopting electronic procurement platforms—to reduce the opportunities for bid rigging. Empirical studies indicate that such measures, when effectively enforced, lead to lower procurement costs and more competitive outcomes (Bosio et al., 2022).

Looking ahead, recent innovations in machine learning and network analysis are poised to further enhance collusion detection. These advanced methods can process vast quantities of bidding data to identify subtle patterns that traditional econometric screens may overlook, offering a more comprehensive toolkit for uncovering collusive behavior. As digital procurement continues to expand globally, integrating these tools into routine oversight could significantly improve market competitiveness and safeguard public resources.

In sum, the post-2000 literature paints a comprehensive picture of collusion in procurement: it is grounded in robust theoretical frameworks, confirmed by empirical evidence across diverse jurisdictions, and addressed by increasingly sophisticated detection and regulatory strategies. This body of work not only deepens our understanding of collusive behavior but also provides practical guidance for interventions aimed at preserving competitive procurement markets.

3. Institutional background

Public procurement in the state of São Paulo is conducted through a decentralized system of purchasing agencies, each referred to as a Public Buyer Unit (PBU). Since 2007, the use of the state's electronic procurement platform, the *Bolsa Eletrônica de Compras* (BEC), has been mandatory for PBUs when acquiring common goods and services. The BEC platform centralizes tender procedures and records detailed information on each procurement process. It plays a crucial role in São Paulo's procurement operations: for instance, in 2018 alone, approximately US$1.7 billion worth of goods and services were procured via BEC. Over the period from 2009 onward, the BEC system handled more than R$7.4 billion (approximately 34% of all state government purchases) for the State

Health Secretariat (SES/SP) and other agencies in São Paulo. Given the large scale and value of these tenders, ensuring fairness and preventing collusion in the procurement process is of paramount importance. A failed tender (one that ends with no supplier selected) not only delays public service delivery but also wastes significant administrative resources, as each tender process can cost the government between US$500 and US$5,200 in preparation and execution.

The BEC catalog contains a comprehensive listing of standardized items and services, classified by category. Nearly 55,000 purchase offers were issued through BEC during our study period, covering about 6,350 distinct items ranging from medical supplies to office equipment. Each purchase offer (PO) is a digital solicitation document that identifies one or more items to be procured, along with the required quantities and specifications. Each item within a PO is auctioned separately; thus a unique procurement transaction can be defined by a combination of a PO number and an item code (POI). The BEC records include rich information for each POI, including the reference (reserve) price set by the buyer, the number of participating firms, all bids submitted (identifying winners and losers), the auction format used, and the identities of the buyer (PBU) and auctioneer.

São Paulo's PBUs predominantly use two types of competitive tendering procedures available on BEC: (i) a sealed-bid tender (*convite*) and (ii) a multi-round descending auction (*pregão*). In a sealed-bid tender, firms submit confidential bids by a specified deadline. After the submission period closes, all bids are opened and the contract is awarded to the qualified bidder with the lowest bid price. The sealed-bid format is typically used for purchases up to a value limit of R$176,000 (approximately US$35,000). In contrast, the descending auction (pregão) has no value limit and involves an open, interactive bidding phase. Initially, firms submit sealed proposals which are evaluated for qualification. Then, all qualified bids are announced (with the bidders' identities concealed) and an open auction phase commences. During this phase, which lasts for an initial 20 minutes, bidders can submit progressively lower bids, knowing the current lowest bid in real time. If a new bid is placed in the final minutes of this period (between 16 and 20 minutes), the auction is automatically extended by an additional 4 minutes. The auction concludes when no new bids arrive for 4 consecutive minutes. The winner is the bidder offering the lowest price at the end of this process, provided that price is below the government's preset reference price.

The key difference between the sealed-bid and descending auction formats lies in the opportunity for iterative bidding and post-auction negotiation. The pregão format,

after determining the lowest bid through the auction, often allows a brief negotiation or confirmation step to potentially further reduce the price or clarify terms with the lowest bidder, as permitted by Brazilian procurement rules. In contrast, the convite (sealed bid) format is a single-shot process with no negotiation stage, which makes it simpler but also means the outcome is entirely determined at bid opening. From an oversight perspective, the open auction format provides more transparency during the bidding process (as all bids are observable in real time, albeit anonymously), whereas the sealed bid format concentrates all transparency at the moment of opening. Both mechanisms are subject to anti-collusion regulations and audit processes, but their procedural differences can influence how collusion or favoritism might manifest. Our analysis will focus specifically on sealed bid auctions.

4. Identifying anti-competitive behaviors

Our empirical strategy follows the approach of Kawai et al. (2023) to detect subtle signs of collusion or favoritism in auction outcomes. The core idea is to focus on procurement auctions where the competition is extremely tight, meaning the winner's bid is nearly identical to the runner-up's bid. Under genuinely competitive conditions, two firms bidding almost the same price should be similar in their cost efficiency and should not systematically differ in their past bidding success. Thus, absent any collusion or external favoritism, one would expect no abrupt differences in firm characteristics when moving from just losing an auction to just winning it.

We implement this idea using a regression discontinuity design (RDD) centered around the margin of victory. Let $\Delta_{i,t}$ be the difference between firm i's bid and the lowest competing bid in auction t. This margin $\Delta_{i,t}$ serves as our running variable, with a cutoff at zero. Firms with $\Delta_{i,t} < 0$ win the auction (their bid is lower than the best alternative), whereas those with $\Delta_{i,t} > 0$ lose. Intuitively, $\Delta_{i,t} = 0$ represents a threshold between winning and losing. We focus on auctions where $|\Delta_{i,t}|$ is very small, meaning the winner barely underbid the closest rival. In such cases, any systematic jump in firm outcomes at $\Delta_{i,t} = 0$ can be interpreted as evidence of an outside influence on the auction result.

The two primary outcomes we examine are the firm's *incumbency status* and its bid timing, identifying whether it is the *last bidder* in the auction. We define an *incumbent* as a firm that won the last auction it participated in prior to the current auction (in-

cumbency is an indicator variable equal to 1 if the firm was the previous winner and 0 otherwise). *Last bidder* identifies whether a firm was the last among the competitors to submit the bid. By examining these variables for firms just on either side of the winning threshold, we can detect irregular patterns: for instance, an unexpectedly high incumbency rate or unusually larger probability of being the last bidder for firms that barely win compared to those that barely lose.

Formally, we estimate the discontinuity at $\Delta = 0$ for these outcomes. Let $x_{i,t}$ denote a generic outcome (such as the incumbency indicator or backlog) for firm i before auction t. The discontinuity is captured by the difference in the expected value of $x_{i,t}$ immediately to the left and right of the cutoff:

$$\beta \;=\; \lim_{\epsilon \to 0^+} \mathbb{E}[\, x_{i,t} \mid \Delta_{i,t} = \epsilon \,] \;-\; \lim_{\epsilon \to 0^-} \mathbb{E}[\, x_{i,t} \mid \Delta_{i,t} = \epsilon \,] \;,$$

where ϵ is a small positive increment in the bid margin. In the absence of any strategic behavior or external interference, we would expect $\beta \approx 0$ for characteristics like incumbency, since firms just winning or just losing should be comparable. Conversely, a significant $\beta \neq 0$ indicates a discontinuous jump in firm characteristics at the threshold, consistent with non-competitive forces affecting the outcome.

We hypothesize that a positive advantage conferred to narrowly winning firms would manifest as $\beta > 0$ in our setup (because the outcome for winners exceeds that for losers). Detecting a substantial gap in these measures at $\Delta = 0$ will allow us to distinguish between a competitive market and one influenced by collusion or favoritism.

For example, if there is political favoritism benefiting certain firms, those firms would appear as incumbents much more often when they win narrowly, whereas otherwise identical firms that narrowly lose would not show such a history. Similarly, a collusive bid-rotation scheme might lead to a more equal distribution of recent wins among participants, in which case we would *not* expect a sharp drop in backlog for losing firms at the threshold.

5. Data

5.1. Sources

We utilize a rich administrative dataset of public procurement auctions conducted via the BEC platform in São Paulo state. The dataset covers the period from January 2009

to December 2019 and includes detailed bid-level information for auctions of common goods and services. During this period, 1,344 PBUs (encompassing state-level executive agencies, the legislative and judicial branches, as well as some municipal governments and even a few private organizations authorized to use the system) carried out procurement through BEC. In total, the data comprise 3,866,407 individual bidding transactions submitted by 19,007 distinct firms, corresponding to 169,607 unique procurement items. Each record in the data identifies the purchase offer (PO) and item, the participating firm, the bid amount, and the auction outcome (win or loss), among other details.

The BEC system's hierarchical catalog organizes items into groups and classes. For example, medical supplies are grouped under a specific category (e.g., group 65 for medical and hospital items), with finer classifications down to individual item codes.[1] This structured coding is useful for analyzing procurement patterns by item type or sector. Our analysis leverages the full breadth of goods and services in the dataset, while controlling for any category-specific effects as needed.

For the purposes of our study, we focus on competitive tenders that resulted in a contract award (i.e., at least one winner and one or more losing bids). We exclude tenders that were canceled or received only a single bid, since those do not provide information on competitive dynamics. This yields a sample of auctions where a meaningful competition took place, which is essential for implementing our regression discontinuity approach.

5.2. Variable definitions

We construct key variables to capture firm outcomes and auction characteristics. First, we define a market such that two firms belong to the same market if they participated at least once in the same auction for the same type of good. This market classification is transitive, resulting in 7,581 distinct markets defined within at least one of our defined time periods.

Following Kawai et al. (2022), we construct an *Incumbent* dummy variable based on this market definition. Specifically, this variable equals 1 if a given firm won the last auction it participated in within the same market before the current auction, and 0 otherwise. This indicator reflects whether the firm entered the current tender as a reigning winner (incumbent) in the specific market or not.

[1] An illustrative item is code 110639, which corresponds to "Furosemide 40mg tablets," classified under class 6531 (a category of pharmaceuticals) within group 65.

Additionally, we define the *Last Bid* dummy variable equal to 1 if the firm was the last to submit a bid in the auction, and 0 otherwise.

Finally, we define the *Margin of Victory*, which serves as the running variable in our RDD design. For each auction, the margin of victory is calculated as the percentage difference between the winning bid and the second-lowest bid, expressed as a fraction of the winning bid. Formally, if b_{win} is the winning bid and $b_{\text{runner-up}}$ is the next lowest bid, the margin is $(b_{\text{runner-up}} - b_{\text{win}})/b_{\text{win}}$. When this margin is very small, the auction outcome was decided by a tiny price difference.

6. Empirical Results

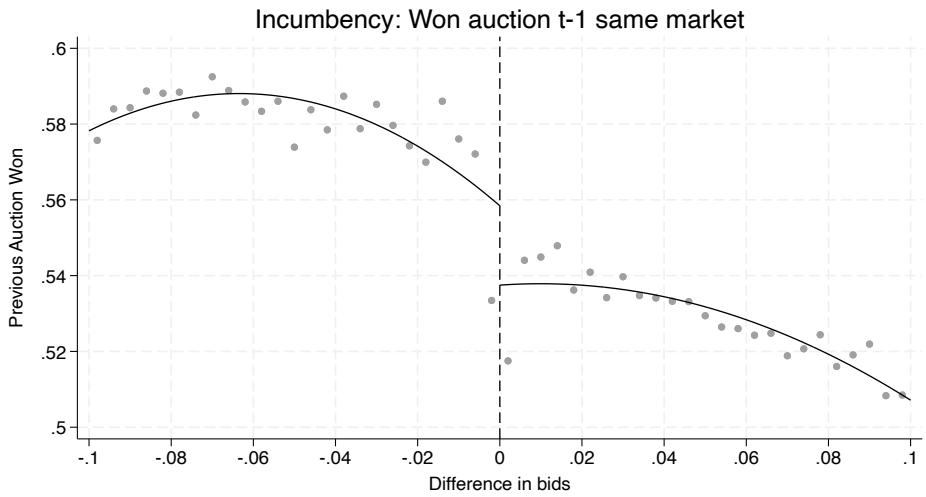
Our empirical analysis identifies statistically significant discontinuities at the threshold of closely contested auctions, suggesting potential anti-competitive dynamics in public procurement markets.

Graphical representations of these effects are illustrated in Figures 1 and 2. Quantitative estimates supporting the graphical evidence are presented in Table 1. Figure 1 shows the estimated probability that the winning firm was the incumbent—defined as the firm winning the immediately preceding auction for a similar good—as a function of the bid margin. A clear discontinuity at the threshold (bid margin equal to zero) indicates that firms narrowly winning auctions (negative bid margin) exhibit a higher probability of incumbency compared to narrowly losing firms (positive bid margin). This pattern is consistent with incumbent firms benefiting disproportionately, which may be reflective of strategic bidding practices or bid rotation schemes.

Similarly, Figure 2 examines the probability that the winning firm submitted the last bid in the auction as a function of the bid margin. A comparable discontinuity at the cutoff suggests firms narrowly winning auctions have a higher likelihood of being the final bidder, potentially indicating strategic bid timing indicative of anti-competitive behavior, such as bid leakage or collusion.

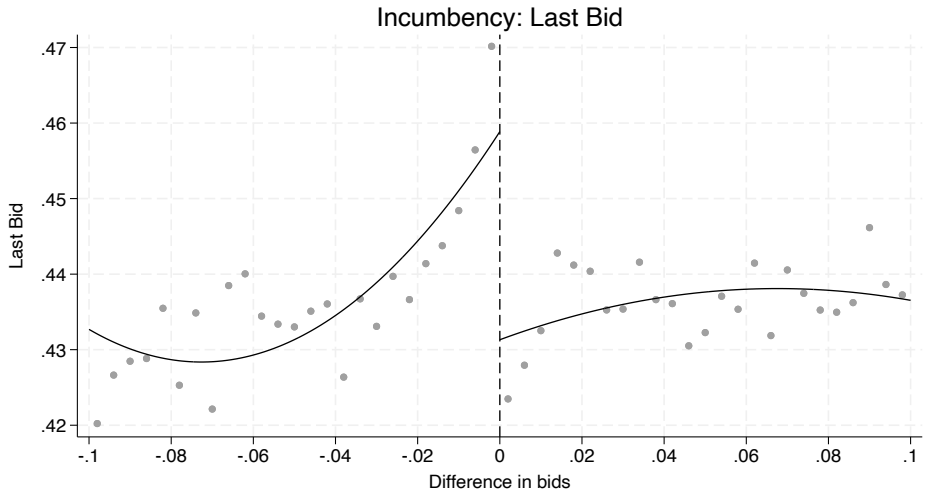
The regression discontinuity estimates corresponding to these graphical results, accounting also for year and market fixed effects, are reported in Table 1. We find that being the incumbent is associated with a 1.82% increase in the probability of winning an auction within the same market. Similarly, winning firms are 6.36% more likely to submit the final bid. Both coefficients are statistically significant at the 1% level. The results remain robust to the inclusion of market and year fixed effects and statistically

Figure 1: Probability of Incumbency around the Bid Margin



Notes: The figure plots binned averages of incumbency status against the bid margin, where negative values correspond to narrowly won auctions and positive values reflect narrowly lost auctions. A distinct discontinuity at the cutoff suggests potential incumbency advantages.

Figure 2: Probability of Being the Last Bidder around the Bid Margin



Notes: The figure plots binned averages of the probability that the winning firm submitted the last bid against the bid margin, where negative values correspond to narrowly won auctions and positive values reflect narrowly lost auctions. A discontinuity at the threshold indicates potential strategic timing in bid submissions.

support the graphical evidence, reinforcing the interpretation of non-competitive forces at play in closely contested procurement auctions.

Table 1: Effect of Winning Auction on Key Outcomes

	Incumbent		Last Bid	
	(1)	(2)	(3)	(4)
Winning Auction	0.0182***	0.0179***	0.0636***	0.0636***
	(0.005)	(0.005)	(0.008)	(0.008)
Observations	261370	260985	156418	156418
Optimal Bandwidth	0.02	0.02	0.01	0.01
Year FE		✓		✓
Market FE		✓		✓

Notes: Standard errors are reported in parentheses. *** indicates statistical significance at the 1% level. The dependent variables are defined as follows: (1) *Incumbent* is a dummy indicating whether the firm won the previous auction within the same market; (2) *Last Bid* is a dummy variable indicating whether the winning firm submitted the final bid in the auction.

7. Additional Results - TBC

7.1. Heterogenety by market

7.2. Correlation with predicted corruption

7.3. Firms outcomes

8. Conclusion

This paper tests a novel approach to detecting anti-competitive behavior in public procurement, applying it to auction data from São Paulo, Brazil. Inspired by Kawai et al. (2023), we use a regression discontinuity design around narrowly decided sealed-bid auctions to identify significant incumbency advantages and strategic bidding behavior. Specifically, narrowly winning firms are more likely to have won prior auctions and to submit the final bid, indicating favoritism or collusion inconsistent with purely competitive bidding.

These findings highlight vulnerabilities in the Brazilian procurement systems, suggesting that both corruption and firm collusion may distort competition. Finally, our analysis provides a practical tool for procurement oversight bodies to flag potential anti-competitive practices and improve accountability in public contracting.

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A. Appendix